

Alex Jacobs

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Education

University of Waterloo

Sep. 2024 – Expected Jul. 2029

BASc Mechatronics Engineering — Robotics and Automation / cGPA 4.00/4.00

Waterloo, ON

Experience

Mechanical Design Co-op — Quadrupedal & Humanoid Robotics

Sep. 2025 – Dec. 2025

Field AI

Irvine, CA

- Led end-to-end mechanical design of a modular sensing and compute payload for the Unitree G1 humanoid across five iterations in 4 weeks alongside three parallel projects, porting existing hardware onto a new platform.
- Designed enclosures adding 20 kg of lead-shot ballast to a quadruped in 3 days, modelling ~63.5% random packing density to hit target mass within 0.2% error and meeting international competition weight rules.
- Assessed IP54 sealing feasibility for a sensing payload using custom splash-test fixtures, identifying insufficient seal clearances that would require a payload redesign.
- Built a modular sensor tower test rig with adjustable camera mounts and configurable extrinsics, saving hours of reconfiguration and accelerating iteration on LiDAR and camera setups.
- Built a Teensy 4.0 data logger to characterize ultrasonic sensor noise, identifying off-axis beam-spread reflections and informing the team's decision on sensor adoption.

Research Assistant — Mobile Robotics & SLAM

Jan. 2025 – Apr. 2025

UCLA Sensing and Robotics for Infrastructure Lab

Los Angeles, CA

- Benchmarked 5+ open-source SLAM systems (DLIO, LIO-SAM, Coco-LIC, ORB-SLAM) on out-of-domain datasets, characterizing localization and mapping failure modes.
- Deployed SLAM packages on lab workstations, patching source and build dependencies to resolve incompatibilities.
- Built a Gazebo simulation from campus models via a Blender pipeline with Python Gaussian noise injection, improving the realism of simulated sensor data.
- Extended SolidWorks models of an unmanned surface vehicle's sensing kits and designed a 3D printed hydrodynamic fairing, reducing drag around its underwater sensors.

Projects

Airborne Object Tracking System — Co-Founder & Mechanical Lead

Aug. 2025 – Present

Panopsys

Toronto, ON

- Co-founded a three-person venture building a ground-based system that localizes airborne objects in 3D by fusing detections from distributed cameras on a ROS2 architecture.
- Own all mechanical design of the sensor head, camera and antenna mounts, and compute, RF, and power integration on a tripod mast, enhancing extrinsic repeatability and speeding field setup.
- Led first outdoor field deployment at an airshow with zero mechanical failures, capturing ~3h of three-node data across 14 aircraft out to ~5 km.
- Driving v2 sensor head, replacing the 20 mm extrusion frame with one printed structure, freeing sensor placement.

Technical Skills

Design & CAD:	SolidWorks, Onshape, Fusion 360, Blender; enclosures, mounts, tolerancing, cable routing
Fabrication:	FDM 3D printing, lathe, mill, drill press, band saw
Robotics & Software:	ROS2, Gazebo, C++, Python, SLAM evaluation
Electronics & Test:	Teensy, wiring and harnessing, sensor characterization, splash testing, field testing

Airborne Object Tracking System

Panopsys / Co-Founder & Mechanical Lead / Aug. 2025 – Present



Node tracking a CP-140 Aurora during the airshow
test



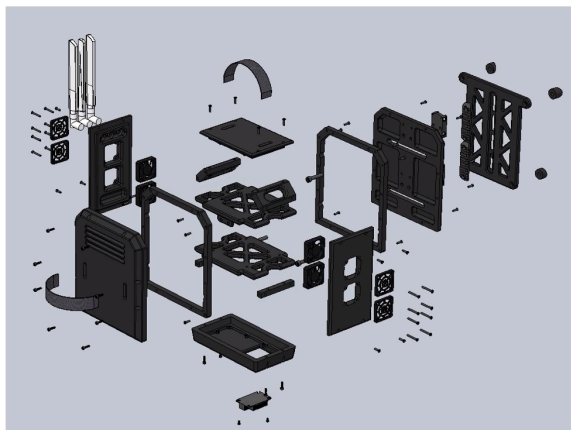
Camera module and radio hardware on the mast

First field deployment: 3 camera nodes | ~100 m baselines | 14 display acts | overhead to ~5 km | ~3 h capture

- Co-founded a three-person venture developing a ground-based system that locates airborne objects in 3D by **fusing detections from distributed cameras**, running on a node-based ROS2 architecture.
- Own all **mechanical design**: camera housings and mounts, antenna placement, sensor-head structure, and integration of compute, RF, and power hardware on a portable tripod mast.
- Designed the head for **repeatable extrinsics** and fast field setup, allowing cameras to be swapped or re-aimed between tests.
- Studied **Kalman filter** theory and implementation to review RTK-GNSS and IMU fusion (ROS2 UKF node) modelling decisions with the software lead.
- Led the first **outdoor field deployment** at an airshow with zero mechanical failures, surfacing electrical and software issues such as an unseated antenna blocking RTK fix.
- Driving **v2 sensor head**, replacing the 20 mm extrusion frame with a single printed structure, removing the mounting constraints that dictated prototype sensor placement.

Back-Mounted Sensing and Compute Payload

Field AI / Mechanical Design Co-op / Nov. 2025 – Dec. 2025



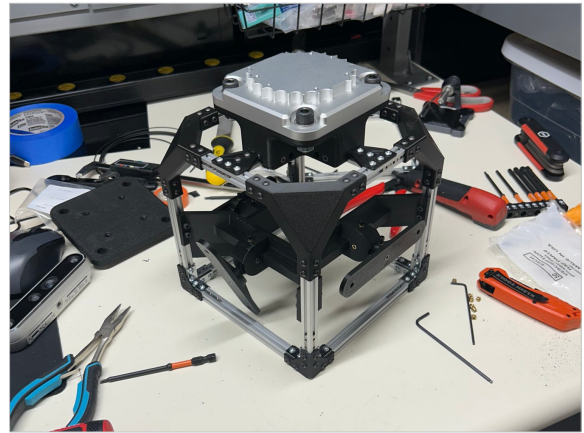
Exploded view of the payload assembly



Fabricated prototypes for cross-office testing

- Conceived and designed the full **payload architecture** for the Unitree G1 from a requirements-only brief, including frame, modular shelf layout, and mounting interface, porting existing hardware onto a new platform.
- Architected internal **cable routing and thermal integration** so existing sensing and compute hardware dropped in cleanly, improving serviceability of connectors and fans.
- Fabricated five **3D-printed iterations** in 4 weeks for Boston office testing, alongside three other projects, and authored design documentation, streamlining handoff to downstream engineers.

- Designed and manufactured a modular **sensor tower** for LiDAR, camera, ultrasonic, and compute payload testing.
- Designed **adjustable camera mounts** with configurable extrinsics in place of the specified fixed design, saving hours of reconfiguration.
- Built fixtures and a **Teensy 4.0** logger to characterize ultrasonic noise, identifying off-axis beam-spread reflections and informing the sensor adoption decision.



Assembled sensor tower test rig

Competition Ballast & Ingress Protection

Field AI / Mechanical Design Co-op / Sep. 2025 – Dec. 2025

- Designed 3 **enclosures** adding 20 kg of lead-shot ballast to a quadraped in 3 days, meeting weight requirements for an international competition in Singapore.
- Sized enclosures for minimum volume using a $\sim 63.5\%$ **random packing density**, achieving target mass within 0.2% error.
- Ran splash-test feasibility checks toward **IP54** for a sensing payload, building custom fixtures for enclosures, fans, and speakers.
- Identified insufficient **seal clearances** that would require a payload redesign; continued as ingress protection point of contact.

Unmanned Surface Vehicle

UCLA Sensing and Robotics for Infrastructure Lab / Aug. 2023, Jan. – Apr. 2025



The unmanned surface vehicle and its sensing kit

- Began as a high school placement in 2023; returned in 2025.
- Extended **SolidWorks** models of above- and below-water sensing kits, including hull components and camera mounts, supporting evolving sensor configurations.
- Modeled and 3D printed a **hydrodynamic fairing** for the underwater sensors, reducing drag.

Simulation and SLAM Evaluation

UCLA Sensing and Robotics for Infrastructure Lab / Jan. – Apr. 2025

- Built a **Gazebo** simulation from campus models via a Blender pipeline with Python Gaussian noise injection, improving sensor realism for a simulated quadraped.
- Deployed SLAM packages on lab workstations, patching **source and build dependencies** to resolve compatibility issues.
- Benchmarked 5+ open-source **SLAM** systems (DLIO, LIO-SAM, Coco-LIC, ORB-SLAM) on out-of-domain datasets, comparing point clouds and localization behaviour to document failure cases.

Design & CAD:

SolidWorks, Onshape, Fusion 360, Blender — enclosures, mounts, tolerancing, exploded documentation

Fabrication:

FDM 3D printing, lathe, mill, drill press, band saw; design for rapid iteration

Robotics & Software:

ROS2, Gazebo, C++, Python, SLAM evaluation

Test & Validation:

Teensy-based sensor characterization, field testing, splash testing toward IP54